Methods survey - Modern methods

Dimensionality reduction

Clustering and segmentation

Statistical models

Latent variable models

Dynamics/state space models

Mechanistic models

Inferring connectivity

Spatiotemporal pattern analysis

Machine learning models

Comparing data to ML models

Variational inference/ELBO

Learning and inference techniques

Information theoretic methods

Dimensionality reduction (I)

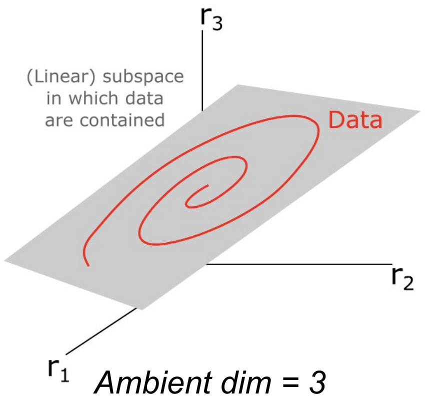
Project or embed data onto “lower-dimensional” subspace or manifold

Often data = population activity patterns, e.g. firing rates estimated by binning spike trains

There are MANY definitions of dimensionality

Three important classes

- Ambient dimensionality: How many numbers per data point in experimental recording
  - Usually determined by experiment setup rather than intrinsic structure in data
- Linear dimensionality: Number of basis vectors needed to span subspace containing data — captures linear structure in data
  - PCA-dimensionality: # of principal components needed to explain X% of total variance
    - Dependent on choice of X
  - Participation ratio: Continuous-valued measure of linear dimensionality
    - Gao et al *bioRxiv* 2017, Recanatesi et al *Patterns* 2022
- Nonlinear dimensionality: How many numbers are needed to specify where a data point lives in the ambient space — captures nonlinear structure in data
  - Many methods to estimate this from data
  - Often called “intrinsic” dimensionality



Ambient dim = 3
Linear dim = 2
Nonlinear dim = 1

Dimensionality reduction (II) - Linear methods

- Principal Components Analysis (PCA)
  - Compute covariance matrix of dataset matrix
  - Perform eigendecomposition
  - Returns list of (orthogonal) eigenvectors sorted by how much variance in the data they explain
  - Project data onto top K eigenvectors to reduce to K dimensions
- Singular Value Decomposition (SVD)
  - Mathematically equivalent to PCA applied to both rows and columns of dataset
- jPCA (Churchland et al, *Nature* 2012)
  - PCA variation for identifying rotational dynamics
  - Requires time-varying data
  - Assumes time derivative is rotational matrix (with purely imaginary eigenvalues) applied to system state
- Canonical Correlation Analysis (CCA)
  - Finds projections of dataset A and dataset B that are maximally correlated
  - Hotelling, *Biometrika* 1936
  - Hardoon et al, *Neur Comput* 2004

Dimensionality reduction (III) - Nonlinear methods

- Multi-dimensional scaling (MDS)
  - Kruskal, Psychometrika 1964
- Papers using MDS in neuroscience:
  - Shepard, *Science* 1987
- Local linear embedding
  - Roweis & Saul, Science 2000
- IsoMAP
  - Tenenbaum et al, Science 2000
- t-SNE
  - van der Maaten & Hinton, JMLR 2008
- Papers using t-SNE
  - Berman et al, *J R Soc Interface* 2014
- UMAP
  - McInnes et al, arXiv 2018
- Gaussian Process Factor Analysis
  - Yu et al, *J. Neurophys.* 2009
- LFADS
  - Pandinarath et al *Nature Methods* 2018
- Papers using LFADS:
  - Ames & Churchland, *eLife* 2019
- Manifold inference of neural dynamics (MIND)
  - Nieh, Schottdorf et al, *Nature* 2021
- CEBRA
  - Schneider et al *Nature Methods* 2023

Clustering and segmentation

Goal:

- Split time-series into underlying “states”, each corresponding to one time of pattern
  - E.g. an animal locomotion signal might be split into “walking”, “still”
- *True* number of states is usually not very well defined
  - Scientist must usually pick a number or resolution of states, often controlled by model parameter
- Can be done with simple methods such as threshold crossing (e.g. “walking” → velocity > v_{min}
- These days more common to identify “hidden states” via statistical model
  - Capture variability of observations via generative model
  - Well-posed to incorporate extensions, “stickiness” of states, priors over state distributions, etc.

<u>Example papers where clustering/segmentation is used to process neural and/or behavior time-series</u>

Kemere et al., Journal of Neurophysiology, 2008, https://doi.org/10.1152/jn.01280.2007
Berman et al., Nature Methods, 2014, https://doi.org/10.1038/nmeth.2975
Wiltschko et al., Neuron, 2015, https://doi.org/10.1016/j.neuron.2015.11.031
Berman et al., PNAS, 2016, https://doi.org/10.1073/pnas.1607601113
Baldassano et al., Neuron, 2017, https://doi.org/10.1016/j.neuron.2017.06.041
Vidaurre et al., PNAS, 2018, https://doi.org/10.1073/pnas.1705120115
Calhoun, Pillow, Murthy 2019, https://www.nature.com/articles/s41593-019-0533-x
Wiltschko et al., Nature Neuroscience, 2020, https://doi.org/10.1038/s41593-020-00706-3
Recanatesi et al., Neuron, 2022, https://doi.org/10.1016/j.neuron.2021.10.011
Markowitz et al., Nature, 2023, https://doi.org/10.1038/s41586-022-05611-2

Statistical models

Probabilistic models of a dataset $\mathcal{D}$ i.e.: $\mathcal{D} \sim P(\mathcal{D}|\ldots)$

(Random processes perspective; see Lec 2: Fundamentals)

Common uses

Inference of hidden variables
Learning of hidden parameters
Predicting unseen/future observations

Common conventions

$\mathcal{D} \sim P(\mathcal{D}|\Theta)$ Data distribution specified by parameters Θ

$\mathcal{D} \sim P(\mathcal{D}|\mathbf{x}, \Theta)$ Data distribution specified by inputs **x** and parameters Θ

$\mathcal{D} \sim P(\mathcal{D}|\mathbf{x}, \Theta, \mathcal{H})$ Data distribution specified by inputs **x**, parameters Θ, and hyperparameters $\mathcal{H}$

Important time-series models for neuroscience

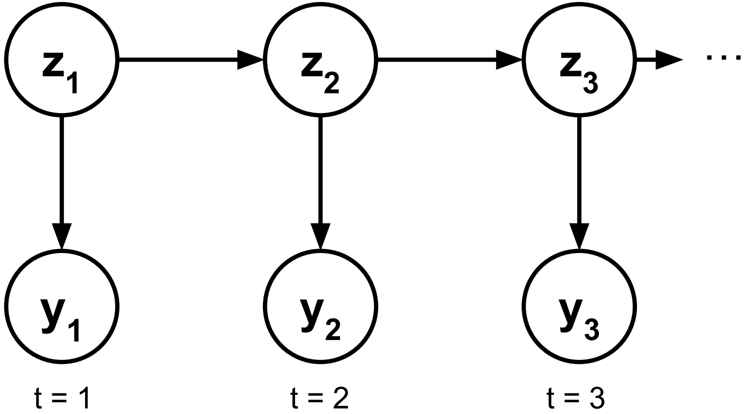
Homogeneous Poisson process: Standard model of spike times given constant firing rate
Inhomogeneous Poisson process: Standard model of spike times given dynamic firing rate
Linear-nonlinear Poisson (LNP) model: Standard model of stimulus-driven spike train
Generalized linear model (GLM): Slightly more general than LNP, since can produce e.g. continuous outputs
Markov Chain (MC): Memoryless model of probabilistic switching among discrete states
Hidden Markov Model (HMM): Hidden variables follow Markov chain, observations are conditioned on hidden variables
Kalman Filter: A continuous version of an HMM, often used for tracking and inference
Variations on Hidden Markov Models: E.g. observations conditioned on hidden variables AND past observations

Graphical models

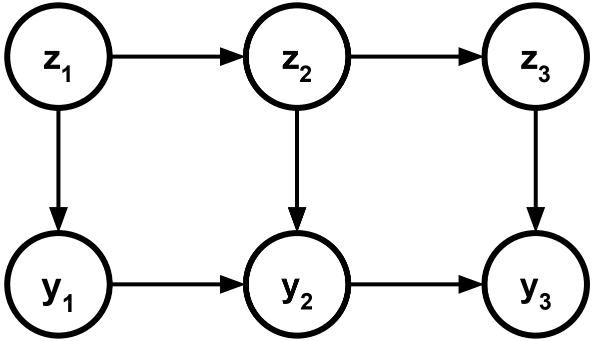
Pictorial summary of joint probability distribution underlying statistical model.
Links (edges) usually indicate conditional probability
Many different conventions for how to interpret links/arrows
For DAG (directed acyclic graphs), specifies how to sample from the distribution.
NOT the same as computation graphs, although can be related.

Common convention:
$\mathbf{z}_t$ → discrete latent
$\mathbf{x}_t$ → continuous latent
$\mathbf{y}_t$ → observed

Example (Hidden Markov Model)



Another example (HMM except each observation also depends on previous observation)



See Bishop 2006, *Pattern Recognition and Machine Learning*, Ch 8: Graphical Models for an in depth review

Latent variable models

Discrete: $$\{\mathbf{y}\} \sim (\{\mathbf{y}\}|\{\mathbf{z}\}, \Theta, \mathcal{H})$$

Example: hidden state **z** represents discrete animal state (engaged vs not engaged)

Continuous: $$\{\mathbf{y}\} \sim (\{\mathbf{y}\}|\{\mathbf{x}\}, \Theta, \mathcal{H})$$

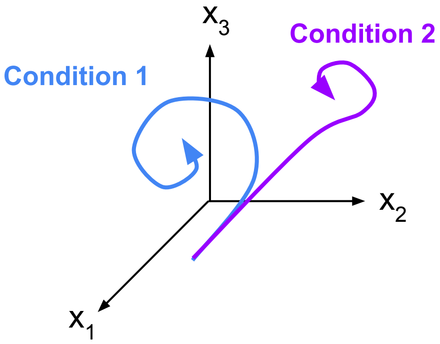
Example: hidden state **x** represents continuous animal state (e.g. arousal level)

Example papers

Smith & Brown, Neural Computation, 2003, https://doi.org/10.1162/089976603765202622
Yu et al., Journal of Neurophysiology, 2009, https://doi.org/10.1152/jn.90941.2008
Pillow et al., Neural Computation, 2010, https://doi.org/10.1162/NECO_a_00058
Fox et al., Annals of Applied Statistics, 2011, https://doi.org/10.1214/10-AOAS395
Linderman et al., ICML, 2017, https://doi.org/10.48550/arXiv.1610.08466
Pandarinath et al., Nature Methods, 2018, https://doi.org/10.1038/s41592-018-0109-9
Ashwood et al 2022 https://www.nature.com/articles/s41593-021-01007-z

State space and latent dynamical systems models

- Enables modeling of time evolution of hidden states underlying observed data.
- Different from imposing statistical model (e.g. Gaussian process) over hidden state time evolution, since state space/LDS usually enforces sequential ("causal") generation of next time points



Latent state **x**(t) evolves via dynamical system

$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, \mathbf{u})$$

External inputs

Observations **y**(t), e.g. neural activity, sampled given **x**

$$\mathbf{y} \sim P(\mathbf{y}|\mathbf{x})$$

- In most studies **x** is lower-dimensional than **y**.
- Many different options to choose for f
  - Linear: well understood but not very expressive
  - RNN: More flexible but harder to understand
  - Piecewise-linear: Middle ground between above
  - Represent f as a neural network ("neural ODE")
- One can also model neural activity directly via dynamical system, rather than via latent dynamics (Rajan et al Neuron 2016)

Papers

Linderman et al, *AISTATS* 2017
Pandarinath et al, *Nat Methods* 2018
Luo et al, *Nature* 2025

Mechanistic models

- Model embodies hypothesis about “how” system works
- “How” usually means specifying a neural circuit, but can be a bit fuzzy
    - “Mechanistic” (e.g. neural circuit) usually contrasted with “Descriptive” (usually refers to model from a general-purpose family of statistical models, e.g. Gaussian process, maximum entropy, etc.) and “Normative” (e.g. set of equations optimized to solve task)

Adapted from Gorur-Shandilya et al, *eLife* 2022 (CC-BY 4.0)

Model of central pattern generator circuit in crustacean stomatogastric ganglion

<u>Ways to gain scientific insight from mechanistic models</u>

- A single model A can recapitulate important phenomena in data → mechanism proposed in model is sufficient

- Model A captures phenomena in data well & better than Model B, Model C, … → suggests mechanism in A is better explanation of data than B & C

- A single model A captures important phenomena well but loses is power when component X is removed → assuming A is a decent model, suggests it relies on X

- Fit model A that captures data well, then do *in silico* experiments to gain insight into system
    - E.g. fit recurrent neural network to neural activity then measure individual currents in model, which you don’t have access to in data; [Perich et al *bioRxiv* 2021])

Connectome-constrained models

- Type of mechanistic model where neural circuit or network can only use connections observed in the connectome
- Various ways connectome can be used as a constraint
    - Only high-level statistics (e.g. mean and variance of degree distribution) used as constraint
    - Cell-type by cell-type statistics used as constraint
    - Connections in model are taken directly from connectome
- Main advantage is that these models force information to flow through real neural pathways
- Once model is fit, scientific insight comes from
    - Identifying key features needed in connectome for good fit (e.g. neuron time constants, which are not included in connectome)
    - In silico studies of model, e.g. analysis of its dynamics in response to inputs, to identify putative mechanisms in brain
    - Predicting experimental outcomes prior to running experiments, in order to increase chance of informative results no matter what

- Caveat: it is often very hard to know functional strength of synapse just from connectome (although usually thought that bigger/more synapses are functionally stronger)
- Usually the exact synaptic weights to be used in the model determined via additional training, using the connectome to mask which connections are allowed to be non-zero

Relevant papers:

Duan, Dong, & Fiete *NeurIPS* 2024
Lappalainen et al *Nature* 2024
Beiran & Litwin-Kumar *Nat Neurosci* 2025
Pugliese et al *biorxiv* 2026

Inferring connectivity

- Start with neural activity, then fit models to this activity to understand how neurons or brain areas are connected

- Can infer *physical connectivity*, i.e. synaptic weights
  - Can then use these weights in a model to investigate in silico or predict experiment outcomes
  - Generally very hard to do, especially from partially observed neural activity
  - Even with complete access to neural activity, can be very hard to infer connectivity

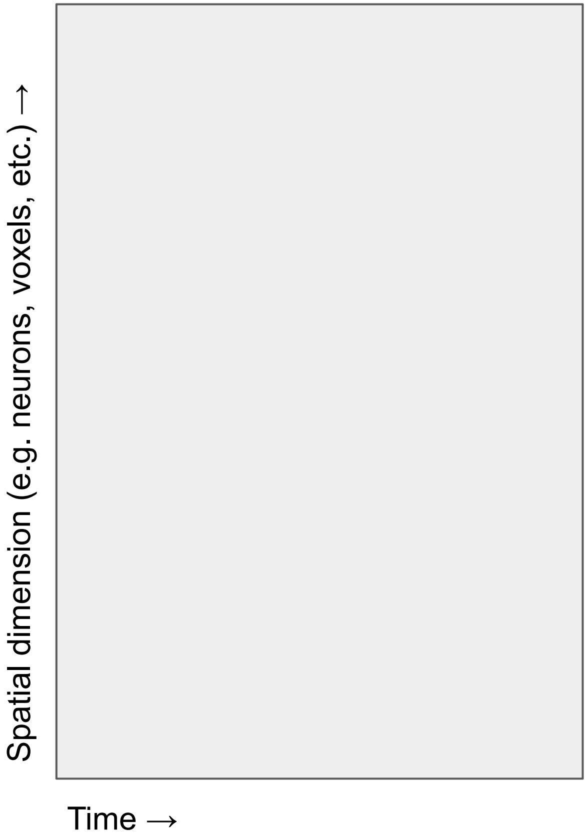
- Or can infer *functional connectivity* to get sense of how information is flowing through brain network
  - Many approaches/definitions of functional connectivity
    - Correlations between neurons or brain area activities
    - Coherence (correlation between frequency bands in different brain areas)
    - Granger causality (nested AR model testing whether hypothetical "causal" area improves predictability)
    - Mutual information (between present activity in one area and past activity in another)
    - Transfer entropy (Mutual information corrected for one area's ability to predict itself)
    - Instrumental variable approach (Optimal perturbations for resolving information flow through network)

Caveats:
- Sometimes activity and connectivity are very well aligned, but sometimes not
- Identifiability issues are significant (e.g. totally different connectivity patterns could fit same activity data)

Some relevant papers: Dhamala et al *Neuroimage* 2008, Stevenson et al *Curr Opin Neurobiol* 2008, Das & Fiete *Nat Neurosci* 2020, Beiran & Litwin-Kumar *Nat Neurosci* 2025

Spatiotemporal analyses

A typical neuroscience dataset



Spatial analyses look for stereotyped “modes” (patterns) in spatial dimension, regardless of timing.
E.g. PCA applied to spatial dimension

Temporal analyses look for “modes” (patterns) in time dimension for individual neurons or pixels
E.g. PCA on temporal stimulus features driving individual neurons

Spatiotemporal analyses look for repeated spatiotemporal patterns, i.e. spatial patterns that unfold in stereotyped sequence

Example papers
Brunton et al *J Neurosci Methods* 2016
MacDowell & Buschman *Curr Biol* 2020

Artificial neural networks - As general-purpose black box models

- ANNs are highly flexible and often well-suited for fitting and predicting time-series data
- Use case is as essentially as general-purpose curve fitting
- Can often outperform simplified or mechanistic models in terms of prediction accuracy
- But are usually considered less scientifically insightful than mechanistic models

- Can use sequential models, e.g. LSTM, in which inputs are fed in one timepoint at a time
- Or feed-forward models that take flattened sequence (e.g. last 10 words) as static input
    - E.g. convolutional neural network, transformer, etc

Ways of using ANNs for scientific tasks:
- Fit black-box model to data to estimate upper bound on predictability
    - Implicitly assumes model is sufficiently capable of finding and using all predictive structure in data
    - Requires due diligence to sweep over architecture and hyperparameters
- Automating easy but tedious task (e.g. identifying all frames in movie where animal is grooming)
- As flexible curve-fitting model to compare which inputs are necessary to produce output
    - E.g. retrain model with different subsets of input given (e.g. past signals up to different time lags) and measure prediction performance to identify what most important inputs are

Caveats:
- Highly expressive models can still be hard/impossible to fit in practice (e.g. vanishing gradients)
- Successful model fit rarely leads directly to scientific insight (always keep scientific question in mind when fitting)

Artificial neural networks - Foundation models

- ANN performance is generally best when trained on very large datasets

- Foundation models pre-train on large datasets and can be fine-tuned on specific datasets
    - Can yield faster training or better performance than training from scratch on a specific dataset

- Generally inherit same advantages and disadvantages of other ANNs
    - Highly flexible and can make good predictions
    - Can be difficult to interpret “how” they work to gain insight into brain mechanisms

Artificial neural networks - As models of the brain

- Neuroscientists are becoming increasingly interested in similarities between AI models and the brain, e.g.
  - Image classification networks explain neural responses to images (Yamins et al PNAS 2014, Pospisil et al eLife 2018)
  - LLMs may represent words and sentence context like humans (e.g. Goldstein et al, *Nat Neurosci* 2022)

- Often amounts to "doing neuroscience" on artificial networks, e.g.
  - Give same inputs to artificial network as to animal and compare animal brain activity vs artificial network response

- Recurrent neural networks
  - Sort-of-brain-like artificial network models
  - Can train them on same tasks as animals, then examine internal dynamics in silico to gain insight into time-varying computation and predict what will be observed in animal brain
  - Or can fit them to data and make

- Example papers where RNNs are used to understand brain dynamics
  - Mante et al *Nature* 201
  - Remington et al *Neuron* 2018
  - Duncker & Sahani 2021 (review)
  - Valente et al *NeurIPS* 2022

Variational inference & ELBO

Technique for computing posterior over latent variable $\mathbf{z}$ given data (observations) $\mathbf{x}$.

Key idea: $p(\mathbf{z}|\mathbf{x})$ is hard to compute so approximate with $q_\theta(\mathbf{z})$, where q is from family of tractable distributions parameterized by θ.

$p(\mathbf{z})$: prior over latents (easy to compute given $\mathbf{z}$)

$p(\mathbf{x}|\mathbf{z})$: likelihood (easy to compute given $\mathbf{x}$ and $\mathbf{z}$)

$p(\mathbf{x},\mathbf{z}) = p(\mathbf{x}|\mathbf{z})p(\mathbf{z})$: joint (easy to compute given $\mathbf{x}$ and $\mathbf{z}$)

Bayes' Rule

$$p(\mathbf{z}|\mathbf{x}) = \frac{p(\mathbf{z})p(\mathbf{x}|\mathbf{z})}{p(\mathbf{x})}$$

Hard to compute

Given data $\mathbf{x}$ and family of functions $q_\theta(\mathbf{z})$, find θ that minimizes $DKL[q_\theta(\mathbf{z}) \,||\, p(\mathbf{z}|\mathbf{x})]$.

Hard to compute

$$DKL[q_\theta(\mathbf{z}) \,||\, p(\mathbf{z}|\mathbf{x})] = \underbrace{E_q[\log q_\theta(\mathbf{z})] - E_q[p(\mathbf{x},\mathbf{z})]}_{-ELBO(\mathbf{z})} + E_q[\log p(\mathbf{x})]$$

To minimize DKL, maximize the ELBO: $E_q[p(\mathbf{x},\mathbf{z})] - E_q[\log q_\theta(\mathbf{z})]$

[Blei et al 2016. Variational Inference: A review for statisticians.]

Simulation-based inference

Inference technique that uses forward simulations instead of calculating likelihood

- For complex models likelihood may be intractable, but can still sample from model via simulation

Useful link between formal Bayesian methods and simulation-driven approaches

Uses artificial neural network to estimate function mapping simulation results to posterior

Basically uses statistical (Bayesian) approach to identify mechanistic models

Papers

Goncalves et al *eLife* 2020: Training deep neural density estimators to identify mechanistic models of neural dynamics
Cranmer, Brehmer, & Louppe *PNAS* 2020: The frontier of Simulation-based Inference
Deistler et al *arXiv* 2025: Simulation-Based Inference: A Practical Guide

Python SBI toolbox from Macke group: https://sbi.readthedocs.io/en/latest/

Popular learning/inference techniques

Empirical Bayes

Bayes

Gradient descent

Expectation maximization

Simulation-based inference

Laplace approximation

Forward/backward
algorithms

Information theoretic methods

Max-ent/max-caliber

Schneidman et al *Nature* 2006
Meshulam et al *Neuron* 2017

Mutual info/transfer entropy

Aguera y Arcas, Fairhall, Bialek *Neur Comput* 2003
Kirst, Timme, Battaglia *Nat Commun* 2016
Fairhall, Shea-Brown, Barreiro *Curr Opin Neurobiol* 2018

Predictive information

Bialek, Nemenman, Tishby *Neur Comput* 1999
Palmer et al *PNAS* 2015

Maximally informative dimensions

Sharpee, Rust, Bialek 2003